Fiscal Year:	FY 2010	**Task Last Updated:**	FY 11/15/2010
PI Name:	Todd, Paul Ph.D.		
Project Title:	Microfluidic Multichannel Flow Cytometer		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **ISSMP**:ISS Medical Project (archival in 2019)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ptodd@techshot.com	**Fax:**	FY 812 923-9598
PI Organization Type:	INDUSTRY	**Phone:**	812 923-9591
Organization Name:	Techshot, Inc.		
PI Address 1:	7200 Highway 150		
PI Address 2:			
PI Web Page:	http://www.techshot.com/		
City:	Greenville	**State:**	IN
Zip Code:	47124-9515	**Congressional District:**	9
Comments:	Note: Company name change to Techshot, Inc. around 2006 (previously Space Hardware Optimization Technology, Inc.).		
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	02/11/2010	**End Date:**	02/10/2012
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Sams, Clarence	**Contact Phone:**	281-483-7160
Contact Email:	clarence.sams-1@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX10CB06C		
Performance Goal No.:			
Performance Goal Text:			
	Techshot, Inc. proposes continued research and development of an on-orbit cell counter culminating in a deliverable hand-held blood cell counter in the form of a "Microfluidic Multichannel Flow Cytometer". Crew health monitoring, the Human Research Program, and research in Fundamental Biology will all benefit from the availability of on-orbit cell counters that can provide immediate and repeated blood counts to assess potential anemia, neutropenia, neutrophila or possibility of infections, and signs of immune cell dysfunction. Three sequential objectives will be completed: (1) Subsystems consisting of a disposable microfluidics chip with multiple channels, fluorescence and light-scatter optics for cell detection, cell labeling protocols and reagents, electronics for controls and data processing and digital analysis and display hardware and software will be constructed and tested on the basis of Phase I research results. (2) Subsystems will be integrated into a benchtop flow cytometer for counting red and white blood cells and three subsets of white blood cells. (3) A compact hand-held cytometer device will be delivered to the sponsor. This handheld device can		

Task Description:	be operated by any on-board personnel and requires only a single drop of blood with fully automated processing of blood on-chip. POTENTIAL NASA COMMERCIAL APPLICATIONS: Sample preservation, especially blood cell samples, on ISS has been a long-standing problem, and a cytometer that can process fresh samples on orbit will solve many problems, especially in space immunology, where knowledge is critical to future space exploration. The immune system of crew members, compromised by weeks of low gravity, renders them more susceptible to acute radiation effects of a solar proton storm. Post-irradiation blood cell counts of irradiated humans has always been vital to their appropriate medical care. Thus a (very compact) blood cytometer will also be critical to future deep space exploration missions. Therefore, infusion into NASA mission needs and projects constitutes providing this technology for (1) cell and immunology research aboard the ISS and ISS National Lab, (2) medical monitoring of ISS crew members via the Human Research Program (HRP), and (3) medical monitoring of deep-space exploration crews, especially following a solar proton storm.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	With its low cost and potential for automated manufacturing the proposed Microfluidic Multichannel Flow Cytometer, in specialized versions can be used for the monitoring of CD4+ T-lymphocyte counts in AIDS patient care in underserved populations, blood cell counting in neonates using a drop of heel-prick blood (National Institute of Child Health and Human Development), in-field blood tests of military personnel, and of research and testing laboratories that cannot afford, or do not have access to, a cytometry facility. A major advantage is that it processes a single drop of whole blood automatically inside the chip allowing it to be used in low-resource settings. The potential market for such a device is large. Many of these customers are identifiable personally by the investigators, and the product can be advertised in cell biology journals and scientific instrument catalogues.
Task Progress:	New project for FY2010. Reporting not required for this SBIR Phase 2 project. (Note: task added to Task Book in November 2010 when received information.)
Bibliography Type:	Description: (Last Updated: 02/27/2004)